NetMind Review 1: Simple Guide (with NS2)

Mohammed Affan (24BAI0014) and Hardik (24BCI0017) | Computer Networks Lab, BCSE308P | For a 6 to 7 minute presentation

Part 1: Understand the project

The problem

A computer network is always producing data: packets, connections, speeds and delays. Tools like Wireshark can capture all of this and show it on screen.

But those tools only **show numbers**. They do not tell you what the numbers mean.

Example: the tool shows "latency = 185 ms". Is that bad? What caused it? What should you check? The tool does not say. Only a person with networking experience can work it out.

So the problem is: monitoring tools give data, but not understanding.

Our solution in one line

NetMind measures the network, detects when something is unusual, and explains it in plain English. Remember three words: Measure, Detect, Explain.

NetMind gets network data from **two sources**: live packets captured from a real network, and a **simulated network built in NS2**. Both go through the same analysis.

How we solve it, step by step

Step	What happens	Tool we use
1. Get data	**Live:** copy the packets passing through the network and read each one's source IP, destination IP, ports, protocol, size and time. **Simulated:** build a network in NS2, create normal traffic, congestion and a link failure, and read the trace file it produces.	Scapy (live), NS2 (simulation)
2. Analyse	Identify the protocol (TCP, UDP, ICMP, DNS, HTTP). Calculate latency, throughput, packet loss and jitter. Build a map of the devices and find the shortest path between them.	Python, NetworkX, Dijkstra's algorithm
3. Detect	A machine learning model looks at the traffic every few seconds and flags it if it looks different from normal.	Isolation Forest (scikit-learn)
4. Store	Save all results with history, so "now" can be compared with "earlier".	SQLite
5. Explain	The user asks a question. The system finds matching networking notes (this search is called RAG), gives the notes and the measurements to an AI model, and the AI writes the explanation.	Sentence Transformers, FAISS or ChromaDB, an LLM
6. Show	Put the metrics, graphs, alerts, network map and the AI chat on one screen.	Streamlit, Plotly

Steps 1, 2 and 4 are the Computer Networks part. Steps 3 and 5 are the AI part, which sits on top. Step 6 displays everything.

Where NS2 fits in

NS2 (Network Simulator 2) is a tool that simulates a network on your computer. You write a short script in a language called Tcl that creates nodes, links, TCP and UDP traffic and router queues. NS2 then runs it and writes a **trace file** (every packet sent, received or dropped) and a **NAM file** (an animation you can watch).

We add NS2 as a new module: **the NS2 Simulation Module**. It gives NetMind three things that a real hotspot cannot:

- **Controlled problems.** We create congestion and a link failure on purpose, at known times. So we can check whether NetMind measures them correctly and whether the ML model flags them.
- **Routing in action.** NS2 has Distance Vector and Link State routing built in. When we break a link, we can watch the traffic take another path.
- **Repeatable results.** The same script always gives the same network, so tests can be repeated.

Our NS2 scenario has 5 nodes and runs for 10 seconds:

Time	What happens	What NetMind should see
0.5 s	TCP file transfer starts from n0 to n3	Normal latency, no loss
3.0 s	A heavy UDP stream from n1 joins on the same 1 Mb link	Queue fills, delay rises, packets dropped: congestion
6.0 s	UDP stream stops	Network recovers
7.0 s	Main link n2 to n3 fails	Distance Vector reroutes traffic through n4, so delay goes up
8.5 s	Link comes back	Traffic returns to the short path

A Python parser reads the NS2 trace file and calculates throughput, packet loss, delay and jitter for each second. These go into the same database and dashboard as live data.

The example to remember

The network suddenly becomes slow. NetMind measures latency of 185 ms, packet loss of 7.4% and many TCP retransmissions. The ML model flags the traffic as unusual. The user asks, "What is causing the slowdown?" The Copilot answers that it is most likely congestion, and suggests checking which flows are using the most traffic. (These numbers are made-up examples from our document, not real measurements.) We will recreate this exact situation in NS2, so it can be shown and tested.

Words you must know

Word	Meaning in one line
Packet	A small piece of data travelling on the network, with a header saying where it is from and where it is going.
Protocol	A set of rules for communication. TCP is reliable. UDP is fast with no guarantee. ICMP is used by ping. DNS turns names into IP addresses. HTTP is web traffic.
Latency	The delay between sending a request and getting the reply.
Throughput	How much data is delivered per second, in Mbps.

Word	Meaning in one line
Packet loss	The percentage of packets that never reach the destination.
Jitter	How much the delay changes from one packet to the next.
Topology	The map of which devices are connected to which. We store it as a graph of nodes and links.
Dijkstra's algorithm	A method to find the shortest path between two nodes in a graph.
Anomaly	Traffic that looks different from normal. It means "worth checking", not always an attack.
Isolation Forest	The ML algorithm we use to find anomalies. It needs no labelled attack data, which is why we chose it.
LLM	Large Language Model. An AI model that reads text and writes text.
RAG	Retrieval-Augmented Generation. Before the LLM answers, we first search our networking notes and hand it the relevant ones, so the answer is based on correct theory.
Copilot	Our name for the AI assistant that answers the user's questions about the network.
NS2	Network Simulator 2. Software that simulates a network so we can create traffic and faults on purpose.
Tcl script	The file (.tcl) where we describe the simulated network: nodes, links, traffic and events.
Trace file	NS2's output (.tr). One line per packet event: + queued, - sent, r received, d dropped.
NAM	Network Animator. Shows the NS2 simulation as an animation: packets moving, queues filling, drops.
DropTail queue	A router queue that drops new packets when it is full. This is how congestion causes loss.

Three points to repeat during the review

- **Networking comes first.** The AI only works on data that the networking modules measure.
- **Two data sources.** Live capture with Scapy and simulation with NS2, both analysed the same way.
- **The AI only advises.** It explains and recommends. It never changes or controls the network.
- **This is a proposal.** Say "we will", not "we have built". All numbers in the document are illustrative.

Part 2: The PPT, slide by slide

The PPT file is NetMind_Review1.pptx: nine slides, about six and a half minutes. Slides 1 to 5 and 8 cover the six items sir asked for: title, team details, abstract, problem statement and objective, architecture and methodology, and expected result. What to say is also in the speaker notes under each slide, so you can see it in Presenter View.

Slide	Title	Speaker	Time
1	Title and team details	Hardik	15 sec
2	Abstract	Hardik	30 sec
3	Problem statement	Hardik	35 sec
4	Objective	Hardik	40 sec
5	Proposed system architecture	Affan	75 sec
6	Methodology: NS2 simulation	Affan	55 sec
7	Example use case	Hardik	40 sec
8	Expected result	Affan	40 sec
9	Plan and work split	Affan, then Hardik	50 sec

Slide 1: Title and team details

Hardik | 15 sec

WHAT IS ON THE SLIDE

- NetMind: AI-Powered Network Intelligence and LLM Network Copilot
- Mohammed Affan (24BAI0014), Hardik (24BCI0017)
- Computer Networks Lab, BCSE308P, Project Review 1

WHAT THIS SLIDE MEANS

Just introduces the project and the team.

> **WHAT TO SAY (Hardik)**
> "Good morning sir. I am Hardik and this is Mohammed Affan. Our project is NetMind, a network monitoring system with an AI layer that explains what is happening in the network."

Slide 2: Abstract

Hardik | 30 sec

WHAT IS ON THE SLIDE

- Three short paragraphs: the problem, what NetMind does (including NS2), and the AI layer
- Right side: keywords (Packet Analysis, TCP/IP, NS2 Simulation, Anomaly Detection, RAG, LLM Copilot)

WHAT THIS SLIDE MEANS

A one-breath summary of the whole project before going into detail.

WHAT TO SAY (Hardik)

"In short, sir: networks generate a lot of traffic and performance data, but monitoring tools only show raw numbers."

"NetMind captures and analyses network traffic, both live and from an NS2 simulation. It identifies protocols and calculates latency, throughput, packet loss and jitter."

"On top of this, a machine learning model detects unusual traffic, and an LLM Copilot explains the problem in simple language. The Copilot only advises and never controls the network."

Slide 3: Problem Statement

Hardik | 35 sec

WHAT IS ON THE SLIDE

- Title: "Problem statement", with the line "Monitoring tools give data, not understanding"
- Left box: what the tool shows (latency 185 ms, packet loss 7.4%, retransmissions, connections)
- Right red box: what the user needs to know (Is this bad? What caused it? Which link? What to check?)

WHAT THIS SLIDE MEANS

Tells sir why the project is needed: tools give data, but not understanding.

WHAT TO SAY (Hardik)

"Sir, modern networks generate a lot of data: packets, connections, latency and throughput. Tools like Wireshark can capture and display all of it. But they only show raw numbers."

"If a user sees high latency or a sudden rise in connections, he still has to work out what it means, what caused it and what to do. That needs expertise. So the problem is that monitoring tools give data, but not understanding."

Slide 4: Objective

Hardik | 40 sec

WHAT IS ON THE SLIDE

- Three numbered cards: 1 Measure, 2 Detect, 3 Explain, each with the tool used
- Under Measure: "Live capture (Scapy) + NS2 simulation"
- Bottom bar: "Real network data is the foundation. The AI only explains it and never controls the network."

WHAT THIS SLIDE MEANS

Gives the whole idea in three steps before the detailed diagram comes.

WHAT TO SAY (Hardik)

"Our objective is to build NetMind, which works in three steps."

"First, measure. We get network data from live packet capture and from an NS2 simulation, and calculate latency, throughput, packet loss and jitter."

"Second, detect. A machine learning model checks whether the traffic is unusual."

"Third, explain. An AI Copilot uses these measurements and a networking knowledge base to explain the problem in simple language and suggest what to check."

"The important point is that real network data is the foundation. The AI only explains it and does not control the network. Affan will now explain the architecture."

Slide 5: Proposed System Architecture

Affan | 75 sec

WHAT IS ON THE SLIDE

- The architecture from Figure 2 of your document, redrawn left to right, with NS2 added
- Two inputs on the left: Live Capture and NS2 Simulation (marked NEW)
- Colour key: navy = networking, orange = AI, grey = storage, teal = dashboard

WHAT THIS SLIDE MEANS

This is the most important slide. It shows every module and how data flows from left (the two inputs) to right (the dashboard). Point at each box as you name it.

WHAT TO SAY (Affan)

"Sir, this is our architecture. It is a pipeline from left to right."

"Data comes in from two sources. Live packets are captured using Scapy, and simulated traffic comes from NS2 as a trace file. The Traffic Analyzer then identifies the protocol: TCP, UDP, ICMP, DNS or HTTP."

"Three modules work on this traffic. The Performance Analyzer calculates latency, throughput and packet loss. The Topology module builds a graph of the nodes and finds shortest paths using Dijkstra's algorithm. The ML module uses Isolation Forest to detect anomalies."

"All results are stored in a database. When the user asks a question, the RAG engine finds the relevant networking knowledge, and the LLM Copilot combines it with the current network state to give an explanation."

"Finally, everything is shown on the dashboard. Each module can be developed and tested separately. I will now explain the NS2 module."

Slide 6: Methodology: NS2 Simulation

Affan | 55 sec

WHAT IS ON THE SLIDE

- Left: the 5-node network. n0 and n1 send into router n2, the red line n2 to n3 is the 1 Mb bottleneck, the dashed path through n4 is the backup
- Right: the timeline (0.5 s, 3.0 s, 6.0 s, 7.0 s, 8.5 s). Red rows are the problems we create
- Bottom: netmind_scenario.tcl, NS2 runs, trace file, parse_trace.py, metrics

WHAT THIS SLIDE MEANS

Shows how NS2 is used: to create known network problems and check that NetMind detects and explains them. It also shows routing (Distance Vector) working, which is a syllabus topic.

WHAT TO SAY (Affan)

"Sir, as per your suggestion, we have added an NS2 Simulation Module."

"In NS2 we build a five-node network. A TCP file transfer runs from n0 to n3. At 3 seconds, a heavy UDP stream joins on the same 1 megabit link. The router queue fills up, delay increases and packets get dropped. This is congestion."

"At 7 seconds we make the main link fail. Since we use Distance Vector routing, traffic is rerouted through the backup node n4, and the delay goes up."

"NS2 writes every packet event into a trace file. Our Python parser reads it and calculates throughput, packet loss, delay and jitter for every second. Because we know exactly when we created each problem, we can check whether NetMind detects and explains it correctly."

Slide 7: Example Use Case

Hardik | 40 sec

WHAT IS ON THE SLIDE

- Left: four numbered steps: Measure, Detect, Ask, Explain
- Right: the dashboard mockup from Figure 5 of your document
- Small caption: "Illustrative values, not measured results"

WHAT THIS SLIDE MEANS

Shows how all the modules work together on one real-looking situation, and what the final screen will look like.

WHAT TO SAY (Hardik)

"Here is an example of how it works. These are illustrative values, not measured results."

"Suppose the network becomes slow. NetMind measures a latency of 185 milliseconds, packet loss of 7.4 percent and a high number of TCP retransmissions. The ML module flags this traffic as anomalous."

"The user asks the Copilot: what is causing the slowdown? The Copilot explains that the likely cause is congestion, and suggests checking the busiest flows and the link utilization."

"We will recreate this same scenario in NS2 to test it. All of this appears on one dashboard, as in this mockup."

Slide 8: Expected Result

Affan | 40 sec

WHAT IS ON THE SLIDE

- Three columns: Networking, NS2 simulation, Intelligence, each with three results
- Bottom bar: "Final result: one dashboard showing live or simulated network state, alerts and AI explanations"

WHAT THIS SLIDE MEANS

Tells sir what will exist at the end of the project. Sir asked for this item specifically.

WHAT TO SAY (Affan)

"Sir, at the end of the project we expect three kinds of results."

"On the networking side, NetMind will capture live traffic, classify it by protocol, measure latency, throughput, loss and jitter, and show the topology with shortest paths."

"With NS2, we will create congestion and a link failure on purpose, show Distance Vector rerouting in NAM, and calculate the same metrics from the trace files."

"On the intelligence side, the ML model will flag unusual traffic and the Copilot will explain the cause and suggest what to check."

"The final result is one dashboard that shows all of this together."

Slide 9: Plan and Work Split

Affan, then Hardik | 50 sec

WHAT IS ON THE SLIDE

- Left: three big numbers: 24 syllabus topics, 8 phases, 2 data sources
- Two cards: Hardik's modules and Mohammed Affan's modules
- Bottom: "Together: integration, testing, documentation" and "Thank you. Questions?"

WHAT THIS SLIDE MEANS

Shows the project fits the course, the order you will build it in, and who does what. Both of you have networking work, which matters in a Computer Networks lab.

WHAT TO SAY (Affan)

"Sir, the project maps to 24 topics of our Computer Networks syllabus, such as TCP, UDP, DNS, IPv4 addressing, routing and performance analysis."

"We will build it in eight phases. We start with the networking modules, and add the AI modules after that foundation works."

"NS2 also lets us show congestion control and Distance Vector routing from the syllabus in action."

"For the work split, I will handle the NS2 simulation, the Performance Analyzer, anomaly detection, the RAG engine and the Copilot."

WHAT TO SAY (Hardik)

"I will handle packet capture, the Traffic Analyzer, topology and routing, and the database, and I will lead the dashboard. Integration and testing we will do together."

WHAT TO SAY (Affan)

"Thank you sir. We are happy to take questions."

If sir asks questions

Question	Short answer
Is this a networks project or an AI project?	Networks. Capture, protocol analysis, performance and routing are the core, and the first three phases are only networking. The AI just explains the measured data.
How is it different from Wireshark?	Wireshark shows packets and the user interprets them. NetMind also calculates the metrics, flags unusual traffic and explains the likely cause.
How will you measure latency?	By timing a request and its reply, for example a TCP SYN and its SYN-ACK, or a ping request and its reply.
How will you measure packet loss?	We estimate it from TCP retransmissions, and we can confirm it by sending pings and counting missing replies.
Why Isolation Forest?	It is unsupervised, so it needs no labelled attack data, and it is light enough for a first version.
Can the AI change the network?	No. It only explains and recommends. A human decides what to do.
Where will you capture traffic?	On our own laptops and our own hotspot, where we have permission. For controlled tests we use NS2.
Why use NS2 if you already capture real traffic?	On a real network we cannot create congestion or a link failure on purpose. In NS2 we can, at known times, so we can check that NetMind measures and detects them correctly.
What is in your NS2 script?	5 nodes, a TCP/FTP flow and a UDP/CBR flow, a 1 Mb bottleneck link with a DropTail queue of 10 packets, Distance Vector routing, and a link failure at 7 seconds.
How does NS2 data get into NetMind?	NS2 writes a trace file. Our Python parser reads it and calculates throughput, loss, delay and jitter, which go into the same database as live data.
How do you calculate delay from the trace?	For each packet, the time it reaches the destination node (r event) minus the time it left the source (+ event).
Why NS2 and not NS3?	NS2 was recommended for our lab, and it includes NAM for visual animation. The parser design would work for NS3 too.

If you do not know an answer, say: "Sir, we have not finalised that yet. We will include it in the next review." Do not guess.

How to run the NS2 part (Ubuntu or WSL)

Files: **netmind_scenario.tcl** (the NS2 network) and **parse_trace.py** (turns the trace into metrics).

```
sudo apt install ns2 nam
ns netmind_scenario.tcl
nam netmind.nam
python3 parse_trace.py netmind.tr
```

The first command installs NS2. The second runs the simulation and creates netmind.tr and netmind.nam. The third opens the animation. The last one prints a table of throughput, loss, delay and jitter per second and saves it as netmind_metrics.csv. In that table, look for higher delay and loss between 3 and 6 seconds (congestion) and higher delay between 7 and 8.5 seconds (rerouting).